BUSINESS USE CASE

Cloud waste is a reasoning problem

An evidence-led OCI tenancy audit for platform and FinOps teams.

THE PROBLEM

A live cloud tenancy, CPU metrics, Terraform declarations and policy/pricing files each tell part of the story. A simple low-CPU rule can mistake a month-end batch machine for a disposable idle server. **False confidence costs trust; missed ownership drift costs money.**

WHO USES IT

A cloud platform or FinOps engineer inheriting an OCI tenancy and asking: **What is unused, what is mis-sized, and what exists without an accountable declaration or owner?**

WHAT THE AGENT DELIVERS

01 Cost-waste candidates: idle or mis-sized compute, detached volumes and unassigned reserved IPs.

02 Unattributed infrastructure: live resources missing from Terraform or clear ownership.

03 An auditable decision: accepted, rejected, revised or undetermined—with evidence and reasons.

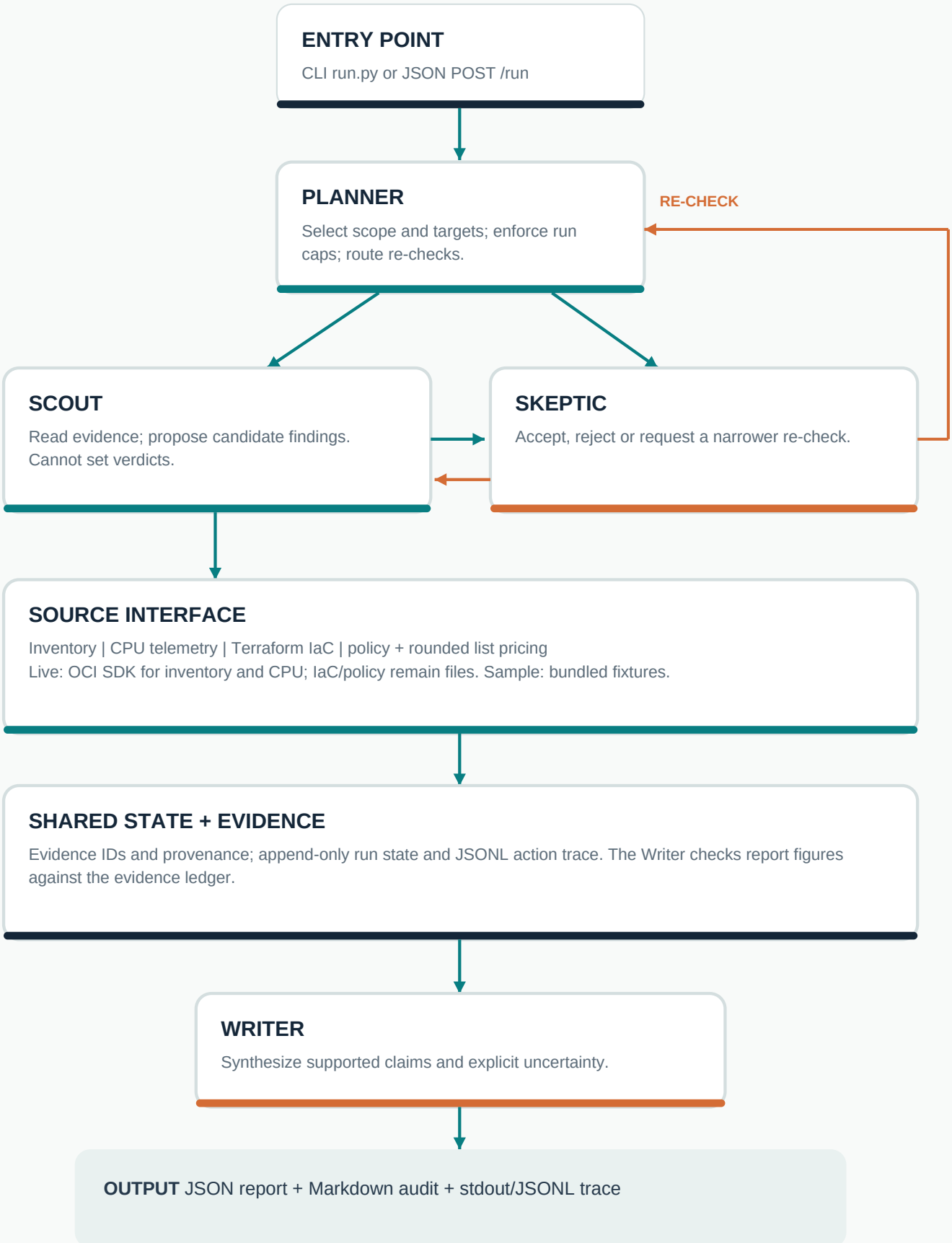
BUSINESS OUTCOME

A prioritized, read-only investigation rather than an automated deletion list. Estimated monthly impact helps triage, but the report preserves uncertainty and explains why a first finding was withdrawn.

SYSTEM ARCHITECTURE

Four agents, one evidence trail

The same bounded investigation runs with fixture data or live OCI adapters.



INVESTIGATION IN ACTION

A finding the agent withdraws

Canonical sample case: batch-01, a machine with a month-end workload.

1

Scout sees an idle machine

Seven days of CPU data: mean 0.766%, peak 0.888%. It proposes an idle-compute cost finding of $90.52/month.

2

Skeptic challenges the premise

The Terraform declaration signals a month-end batch schedule. A seven-day window may miss the job, so the Skeptic asks for a 31-day re-check instead of accepting.

3

Scout revises; Writer preserves history

The wider view exposes a 58.0% peak. The finding changes from “idle” to “mis-sized around a monthly burst.” The old C-001 is rejected and superseded by C-005.

WHY THIS MATTERS

Low utilization alone is not proof of waste. The critic changes the investigation, not merely the wording of the answer. Both the superseded claim and the reason for withdrawal remain visible in the report and trace.

Result in the committed sample: 6 resources examined; 4 accepted findings, 1 rejected and 1 undetermined. Audited list-price spend: $196.18/month; potential waste: $105.66/month, with the revised batch impact explicitly an upper bound.

SOLUTION AND PROOF

Useful, bounded, honest

What is working today, what was evaluated, and what the report cannot promise.

TECHNICAL CHOICES

Evidence before prose. Code computes costs, thresholds, coverage, IaC matching and verdict eligibility. A figure guard refuses unsupported report numbers; model-written prose can be rewritten from rule templates. **Bounded work.** Up to 3 rounds, 25 resources, 2 re-checks per finding; 300 s soft and 540 s hard wall-clock budgets. **Degradation without invention.** A failed source gets one retry; fallback and missing evidence are identified. No CPU verdict is made from substitute telemetry.

PROOF IN THE REPOSITORY

3 curated scenarios

False positive and revision; clean tenancy with zero findings; exceptions, missing coverage and ownership drift.

6 generated tenancies

Committed held-out score file reports zero failures; scored positives have precision, recall and verdict accuracy of 1.0.

Live and clean-run checks

OCI inventory/Monitoring adapters were run against the author’s lab tenancy; credential-free clone and x86/arm64 container paths were verified.

LIMITS AND DISCLOSURES

Read-only recommendations, no remediation. Scope is compute, block volumes and reserved public IPs; CPU is the only metric. Estimated costs use a bundled rounded list-price catalog, not an invoice or guaranteed savings. Live IaC and policy remain file-backed; early lab CPU history was not mature for a full seven-day verdict. Sample mode uses synthetic fixtures and a deterministic rules generator, not a live model. No sponsor technology is mounted. Track spelling and partner stack still require HackOS confirmation before submission.

TRY IT / INSPECT THE EVIDENCE

```
python run.py --scenario data/scenarios/example_1
python -m evals.run_eval
```

Sources: README.md; app/loop.py, app/agents/, app/sources/; output_examples/example_1_report.md, eval.txt, held_out.txt; logs/samples/trace-example_1.jsonl.